

Spoken Tutorial on Building with AI: Understanding APIs

Module	An Introduction to Building with AI
Episode	3. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is. 2. Explain how an API acts as a messenger. 3. Identify the purpose of an API key. 4. Apply the concept of an API request.
Approx. Duration	3-4 min
Outline	• What is an API? • What is an API Key? • Making an API Request • Vague API Request • Improved API Request • Comparing API Request Results • APIs in Real Life: A Grocery App
Meta Tags	API, AI, Artificial Intelligence, API Key, API Request, Building with AI, Spoken Tutorial, Programming, Integration, Messenger, Prompting, Model
Pre-requisite Tutorial	Module 2:Art of the Prompt

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on Building with AI: Understanding APIs.
Learning Objectives Slide	In this tutorial, you will learn to, • Define what an API is. • Explain how an API acts as a messenger. • Identify the purpose of an API key. • Apply the concept of an API request.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should complete Module 2: Art of the Prompt.

Illustration showing API as a messenger between two programs, with a waiter serving food	What if I told you that different computer programs can talk to each other? An API, or Application Programming Interface, makes this possible. It acts like a messenger. Think of an API like a waiter in a restaurant. You give your order to the waiter. The waiter takes it to the kitchen. The kitchen prepares your food. The waiter then brings your food back to you. Similarly, an API takes your request to a service and brings back the response.
Illustration of an API key as an ID card unlocking access to a service	How does a service know it's you making the request? This is where an API key comes in. An API key is a unique secret code. It identifies you and proves you are allowed to access a service. Imagine it as your ID card to enter a special club. Only members with an ID card can place orders. Always keep your API key private and secure.
Person using a grocery app on a phone, typing 'tomato' in search bar	You now know what an API is and how an API key works. Let us see how to make an API request. An API request is like placing an order to a service. It includes your API key for authentication. It also includes your instruction or prompt. The API then sends this request to the model. The model processes your instruction. Finally, the API returns the output to you. Let's see this in action with a grocery app example. Open any grocery app on your phone. Tap the search bar at the top. Type 'tomato' and press Enter. The app sends an API request to the server. The request asks for products matching 'tomato'. The server processes this request. It sends back a list of matching products. Then, the app displays these products on your screen.
Screen close-up: user typing 'Tell me about cars...'	Now, let us try making different types of API requests. First, consider a vague API request. This type of request asks for general information. It lacks specific details. For example, asking an AI model, 'Tell me about cars.' This will likely give you a broad, unspecific response. It will not be very helpful.

Screen close-up: user typing 'Summarize key features of electric cars in 3 bullet points...'	Let's see how to make a better API request. An improved API request is precise. It provides clear context. For example, you could ask to summarize key features of electric cars in 3 bullet points. Focus on their environmental impact and performance. This guides the AI model. It provides a targeted and useful response.
Side-by-side: vague prompt output vs improved prompt output	Let us compare the results of these two requests. The vague request gave a generic overview. The improved request provided specific, actionable information. It was relevant to what we needed. Clear and precise prompts lead to much better AI outputs. Understanding this helps you get the most from AI tools.
Seamless interaction of a grocery app showing 'tomato' search results, powered by APIs	You just saw how an improved request works. Let us revisit our grocery app example. When you search for an item, say 'tomato': The app sends an API request to the server. The server processes your request. It fetches data from its database. It then sends this data back to your app. Finally, the products appear on your screen. This seamless interaction is entirely powered by APIs.
Summary Slide	Let us summarize what we learned. We defined APIs as messengers. We understood API keys for authentication. We learned how to make effective API requests. We also saw a real-life application example.
Assignment Slide	Now as an assignment, Think of another everyday application. Identify how APIs might be used within it. Describe the request and response flow. Compare the results.
EduPyramids logo	This Spoken Tutorial is brought to you by EduPyramids Educational Services Private Limited, SINE, IIT Bombay. Thank you for joining.